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A Theoretical Framework

In this section, we introduce a theoretical framework to guide our empirical analysis.

In Section A.1, we first introduce a simple partial-equilibrium model of chatbot adoption to interpret the descriptive patterns in Section 1.4. We posit a Roy (1951) style selection model of chatbot adoption, in which employer initiatives may shift both the costs and benefits of adoption. We use the model to analyze how these shifts affect workers' adoption decisions and the work-related benefits they report from using the tools.

In Section A.2, we embed the adoption model within a labor market equilibrium model, in which differentiated firms adopt chatbot initiatives and hire workers to maximize profits. We use this model to guide and interpret the empirical analysis of labor market outcomes in Section 3.

A.1 Adoption and Work

A.1.1 Setup

Worker i derives benefits B_i from using AI chatbots:

$$B_i = \alpha_b + \beta_b \times i, \tag{3}$$

where individuals $i \sim \mathcal{U}([0, 1])$ are drawn uniformly from the unit interval and ordered by their individual benefits, such that $\beta_b \leq 0$. While we assume linear functional forms for analytical simplicity, our predictions also hold under more general monotonicity assumptions.

The worker also incurs a cost C_i of adopting chatbots:

$$C_i = \alpha_c + \beta_c \times i, \tag{4}$$

where $\beta_c \geq 0$ if workers who benefit more from AI also face lower adoption costs.²⁰

²⁰A common finding from RCTs is that AI chatbots tend to yield greater benefits for less experienced (and thus typically younger) workers (Brynjolfsson, Li and Raymond, 2025; Dell'Acqua et al., 2023; Noy and Zhang, 2023). If these workers also find it easier to adopt new tools, then $\beta_c \geq 0$.

We conceptualize employer initiatives $E \in \{0, 1\}$ as shifting workers' costs and benefits of using AI chatbots through the parameters α_c and α_b :

$$\alpha = \alpha_0 + \alpha_1 E. \quad (5)$$

While we focus on a binary initiative (e.g., encouraged use vs. not) for analytical simplicity, our propositions naturally extend to combinations of employer initiatives—such as use policies, enterprise chatbots, and training.

Workers adopt AI chatbots if their perceived benefits exceed the associated adoption costs ($B_i \geq C_i$):

$$A_{ij} = \mathbf{1}[B_{ij} \geq C_{ij}] = \mathbf{1}[\hat{\alpha}_0 + \hat{\alpha}_1 E_j + \hat{\beta} \times i \geq 0], \quad (6)$$

where $\hat{x} = x_b - x_c$ denotes net benefits.

A.1.2 Model Predictions

Figure A.1 illustrates how employer initiatives shift adoption rates and the average benefits reported by workers; formal derivations follow below.

Adoption rates. Table 1, Columns 1-2 show how chatbot adoption rates vary with employer initiatives. In the Roy model outlined above, the optimal adoption rate is given by:

$$i^*(E) = \mathbf{P}[B_{ij} \geq C_{ij}] = -\frac{\hat{\alpha}}{\hat{\beta}}, \quad (7)$$

where $0 < \hat{\alpha} < -\hat{\beta}$, corresponding to the empirically relevant case in which adoption rates lie between 0 and 1.

Equation (7) implies that employer initiatives that raise net benefits ($\hat{\alpha}$) increase adoption, with stronger effects when $\hat{\beta}$ is numerically small—i.e., when workers are relatively homogeneous in their net benefits of using AI chatbots.

In this light, the fact that employer-encouraged use in Table 1 is associated with a

doubling of adoption rates suggests that this initiative is powerful relative to the amount of individual heterogeneity in net benefits.

Benefits for users. Table D.2, Columns 3-6 show how reported benefits among chatbot users vary with employer initiatives. The average benefit reported by users is:

$$\bar{B}(E) = \mathbb{E}[B_i(\alpha) \mid i \leq i^*(\alpha)] = \gamma\alpha_c + (1 - \gamma)\alpha_b, \quad (8)$$

where $\gamma = \frac{\beta_b}{2\beta} \geq 0$. Equation (8) provides two key insights into how employer initiatives relate to reported user benefits.

First, employer initiatives that reduce adoption costs (α_c) *lower* the average benefit among users due to a *selection effect*: lower costs induce adoption by workers with lower idiosyncratic benefits (i.e., higher i). This negative selection effect, γ , is stronger when workers are more homogeneous in their net benefits from using AI chatbots (i.e., when $\hat{\beta}$ is numerically small, so that employer initiatives generate larger increases in adoption; cf. Equation (7)) but more heterogeneous in their gross benefits from the tools (i.e., when β_b is large in absolute value, implying that marginal users have lower idiosyncratic benefits).

Second, employer initiatives that increase benefits (α_b) have two offsetting effects on average reported benefits. On the one hand, a higher α_b directly raises each user's individual benefit B_i . On the other hand, it expands adoption, drawing in users with lower idiosyncratic benefits (i.e., higher i), thereby triggering the same negative selection effect as a reduction in α_c . The net effect of an increase in α_b on the average reported benefits among users $\bar{B}$ is positive if and only if $\gamma < 1$. This condition is more likely to hold when workers are relatively similar in their gross benefits from the tools (i.e., when β_b is small) but differ more in their adoption costs (i.e., when β_c is large).